Auditing parameterisation and bound effects in sequential P-wave velocity and spectral-ratio attenuation tomography of the Middle Magdalena Valley, Colombia

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Abstract

We audit parameterisation and bound effects in sequential P-wave velocity and spectral-ratio attenuation tomography of the Middle Magdalena Valley, Colombia. Servicio Geológico Colombiano data are inverted using shared straight-ray geometry, with attenuation conditioned on recovered velocity. Expanding from 14 to 56 receivers increased accepted differential observations from 4 216 to 52 914, but normalised root-mean-square (RMS) reduction changed only from 12.1 % to 12.6 %. Different starting fields and domains preclude isolating receiver count. An illumination audit found that 46 % of sampled ray-hit cells lay outside the inherited domain. Re-parameterisation increased RMS reduction to 15.5 %; the paired event-bootstrap gain was 2.93 percentage points (95 % interval 2.06 to 4.14), conditional on fixed predictions. A controlled mask–velocity experiment attributes almost all this gain to the mask. Stronger station damping reduced component inflation, although substantial station–structure anticorrelation persisted. Archived predictions were evaluated before model clipping. Forward RMS of the clipped attenuation model was 0.11641 s versus 0.0590 s for the unconstrained solution; raising the ceiling from 300 to 1500 gave 0.10882 s. Both clipped predictions fit worse than zero. Consequently, archived RMS invariance cannot establish bound non-identifiability, and below-ceiling Q values remain provisional. A Benioff surface reconstructed from 879 hypocentres differed from Slab2 by a median absolute 8.2 km. These audits delimit interpretation of regional attenuation maps by requiring component-covariance diagnostics and forward evaluation of the bounded model actually displayed.

Keywords: seismic attenuation, local earthquake tomography, spectral ratio, model parameterisation, null space, Nazca slab, Colombia

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1. Introduction

The northern Andes record one of the most complex subduction settings in South America, where the Nazca, Caribbean and South American plates interact with inherited crustal structures, sedimentary basins and strongly segmented intermediate-depth seismicity (Taboada et al., 2000; van der Hilst and Mann, 1994; Syracuse et al., 2016; Schellart, 2023). In Colombia the Bucaramanga seismic nest and the surrounding intermediate-depth seismicity constitute a dense natural source array, yet the relation between hypocentral patterns, elastic anomalies and plate geometry remains scale dependent (Zarifi et al., 2007; Londoño et al., 2019; Sepúlveda-Jaimes and Cabrera-Zambrano, 2018; Hayes, 2018). Attenuation is in principle a sharper discriminant than velocity for temperature, partial melt and fluid content (Hacker et al., 2003; Mavko et al., 2009; van Keken et al., 2011), which motivates Q imaging as a complement to velocity tomography in this setting.

The MMV is an intermontane basin between the Central and Eastern Cordilleras, bounded and crossed by inherited fault systems and filled by Cenozoic clastic sequences that strongly depress shallow seismic velocities and distort local ray geometry (Mojica and Franco, 1990; Alcárcel and Gómez, 2019). The structural framework includes the Bucaramanga–Santa Marta, La Salina and Eastern Frontal fault systems, which link basin inversion, Andean shortening and dextral deformation of the North Andean block (Mojica and Franco, 1990; Audemard M. et al., 2021; Muller et al., 2010). Earlier work established the regional catalogue, the local magnitude scale and crustal-to-upper-mantle velocity structure from ambient-noise and arrival-time inversion (Ojeda and Havskov, 2001; Poveda et al., 2018; Londoño and Romero, 2017; Londoño et al., 2020; Havskov and Ottemöller, 2010), and more recently regional attenuation estimates (Londoño et al., 2022; Castelblanco Ossa, 2023; Castelblanco et al., 2025). The configuration is unfavourable in a specific way: basin heterogeneity, receiver-side site response and intermediate-depth sources are sampled by one elongated network, so receiver terms and path structure compete for the same observations.

Local earthquake tomography relates arrival-time residuals to three-dimensional structure through a linearised inverse problem (Aki and Lee, 1976; Crosson, 1976; Thurber, 1983; Pavlis and Booker, 1980), and the spectral-ratio method extends this to attenuation by cancelling the common source spectrum between two receivers of a single event (Anderson and Hough, 1984; Quan and Harris, 1997; Safak, 1997; Eberhart-Phillips and Chadwick, 2002). In both cases the recovered structure is conditioned by the parameterisation: unsampled cells are undetermined, and parameters that trade off against one another are individually meaningless even where their combination fits the data (Thurber, 1992; Zhao, 2015; Menke, 2018; Rawlinson and Spakman, 2016). For spectral-ratio Q the trade-off is acute. Station terms enter every observation with opposite sign, so their mean is unconstrained and their differences can absorb any path effect the parameterised volume cannot express (Eberhart-Phillips and Chadwick, 2002; Guo and Thurber, 2021; Gao and Wang, 2016).

The station–structure trade-off and inverse non-uniqueness are established in the literature cited above. Our contribution is to make their audit explicit and reproducible in a sequential, bounded-model workflow: report component covariance on the retained observations; distinguish invariance of an archived residual from non-identifiability of a bound; and evaluate the forward prediction of the actual bounded model before interpreting it. The reference and expanded campaigns illustrate why receiver count alone is

insufficient, although their different starting fields and grids prevent a receiver-only comparison. Paired event resampling quantifies sampling variability of the archived fit, and a controlled mask–velocity experiment isolates the parameterisation effect. A damping sweep and structure-only synthetic tests diagnose remaining coupling. The Middle Magdalena models and a seismicity-defined Benioff surface compared with Slab2 (Hayes, 2018) serve as a case study; the clipped attenuation values remain provisional.

2. Tectonic setting and data

2.1. Seismological data

All data derive from the Servicio Geológico Colombiano national network. The inventory contains 1 114 events with QuakeML phase information and archived waveforms; 1112 have retained P-wave picks and appear in the location map. They yield 12 317 P-wave arrival times for the velocity branch. For the attenuation branch, 12 309 vertical-component records were retrieved, of which 12 091 yielded usable signal and noise spectra after instrument-response deconvolution using SGC StationXML poles and zeros. Pairing within events produced 52 957 differential t^* observations, of which 52 914 passed quality control and enter the inversion. The reference campaign similarly produced 4 221 pairs, of which 4 216 were accepted. Raw and accepted counts are kept distinct throughout.

2.2. Receiver configuration

An inventory of the archived QuakeML shows P-phase picks at 189 stations for these events, whereas the reference configuration defined here used 14; the domain filter discarded 76 % of available picks. Retaining all stations within 300 km yields 56 receivers, reduces the median azimuthal gap per event from 197.4° to 82.8°, and reduces the proportion of events with gap $> 180°$ from 68 % to 5 %. Section 4.1 shows that the reference-to-expanded campaign comparison produces little improvement in the archived linear fit.

3. Method

3.1. Grids and starting models

Both branches in the expanded campaigns use a regular Cartesian grid with spacing 20 km, containing 31 by 32 by 12 nodes along x, y and depth, respectively. Only nodes inside the campaign domain and above the hit threshold are inverted; the full computational grid must therefore be distinguished from the active mask. The reference domain extends from $-74.60°$ to $-72.55°$ longitude, 6.20° to 8.05° latitude and to 180 km depth. It uses 14 receivers, 4 269 P picks and 4 216 accepted spectral pairs on a computational grid of 13 by 9 by 12 nodes, also at nominal 20 km spacing. Its computational x range is -186.32 to 53.68 km, and y range -76.21 to 83.79 km, relative to the common geographic origin. Computational extent and the nominal rectangular inversion domain are distinct ray constraints. Its archived starting velocity field is three-dimensional, with maximum within-layer range 1.05 km s^{-1}; the expanded campaigns instead use the one-dimensional profile below. The reference comparison therefore changes receivers, computational support and starting velocity structure together, and is not a controlled receiver-only

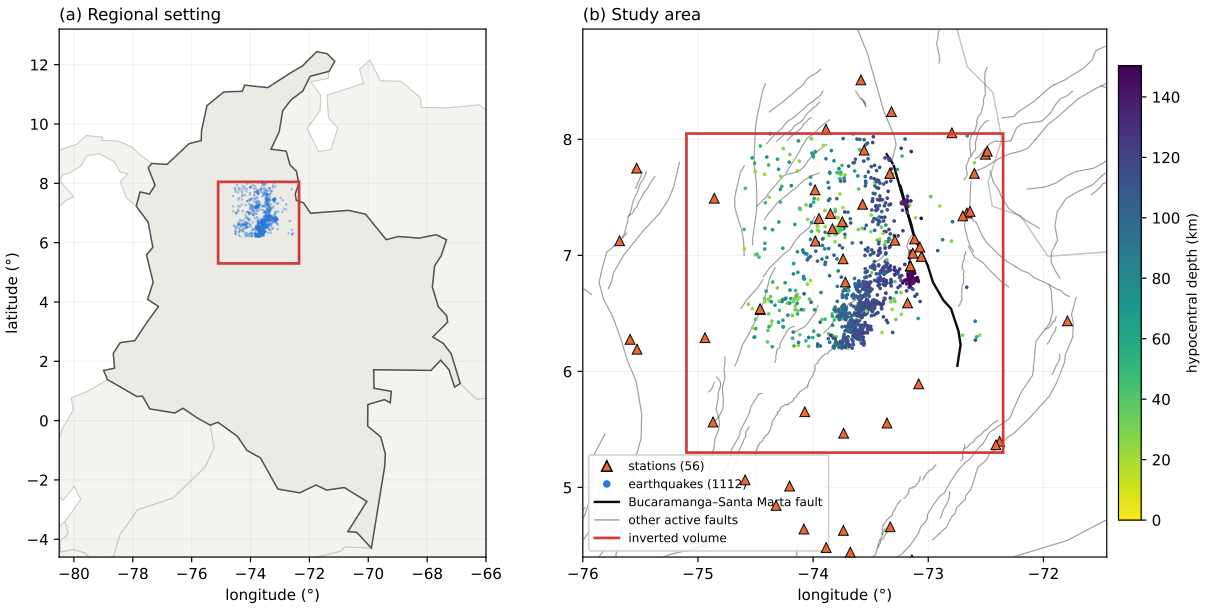


Figure 1: Location of the study area. (a) Regional setting, with the inverted volume outlined in red and the full earthquake set in blue. (b) Enlargement of that volume showing the 1112 earthquakes with retained P-wave picks, coloured by hypocentral depth, the 56 receivers (triangles), and active fault traces from the GEM database (Styron and Pagani, 2020); the Bucaramanga–Santa Marta system is highlighted. The elongation of the permanent network along the basin axis is the geometric origin of the station–structure trade-off analysed in Section 3.4.

test. The complete reference starting field is retained in the model archive. The ray-consistent domain is specified in Section 4.2. The same straight-ray operator, hit thresholds and regularisation weights are used for the reference and expanded comparisons, apart from the declared station damping sweep. In the ray-consistent comparison, changing the domain also changes the recovered velocity field used in the attenuation kernel; this is a sequential effect, not an isolated change of the attenuation mask.

All grids have vertical nodes at 0, 20, 40, 60, 80, 100, 120, 140, 160, 180, 200 and 220 km. Computational storage therefore extends beyond the declared inversion domains. Nodes deeper than the respective domain limits are inactive and retain their starting values. The deepest active velocity and attenuation nodes in both inherited-domain campaigns are at 160 km, because activation additionally requires the hit threshold; in the ray-domain campaigns both branches extend to 140 km. A 220 km storage extent does not imply resolution to that depth.

The expanded campaigns' starting velocity profile is given as depth in km: velocity in km s^{-1}, 0: 5.31, 20: 7.22, 40: 8.03, 60: 8.11, 80: 8.20, 100: 8.28, 120: 8.28, 140: 8.28, 160: 8.28, 180: 8.28, 200: 8.28, 220: 8.28. Initial Q is 40 above 30 km and 60 below. Velocity nodes with at least ten ray hits and attenuation nodes with at least five are active. Velocity smoothing and damping weights are $\lambda_L = 3$ and $\lambda_D = 0.25$; attenuation cell smoothing and damping are 3 and 0.05. The original station damping is 0.15 and the adopted value is 4. These are multipliers of the augmented least-squares rows; their numerical values are tied to the operator units and are not directly transferable to another discretisation.

LSMR uses the implementation's default tolerances atol = btol = 10^{-6} and condition limit conlim = 10^{8}, with at most 1200 iterations. The adopted velocity and attenuation runs terminate after 281 and

328 iterations with stopping code two (least-squares tolerance met). Rays are straight segments sampled every 2.5 km; velocity is solved first and then held fixed in the attenuation kernel. There is no common objective or iterative feedback from attenuation to velocity.

3.2. Velocity branch

Travel-time perturbations are related to slowness perturbations by $\delta t = \sum_k \ell_k \, \delta s_k$, with ℓ_k the path length in cell k (Aki and Lee, 1976; Thurber, 1983; Červený, 2001). The system is solved with LSMR (Fong and Saunders, 2011) under Laplacian smoothing and damping,

$$\min_{\delta \mathbf{s}} \left\| \begin{bmatrix} \mathbf{A} \\ \lambda_L \mathbf{L} \\ \lambda_D \mathbf{I} \end{bmatrix} \delta \mathbf{s} - \begin{bmatrix} \mathbf{d} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \right\|_2^2 . \tag{1}$$

Velocity regularisation weights are fixed across campaigns. The attenuation station damping varies only in the declared sweep. After solving, velocities are clipped to the numerical interval 2–15 km s^{-1}; 4.3 % of active velocity nodes reach the upper limit. This generous upper limit is a numerical safeguard, not a plausible lithospheric velocity or a validated physical constraint. The archived residuals refer to the solution before this clipping.

3.3. Spectral-ratio attenuation branch

For one event recorded at stations i and j, the amplitude-spectral ratio cancels the source term, giving

$$\ln \frac{A_i(f)}{A_j(f)} = c - \pi f \, \Delta t^*_{ij}, \qquad \Delta t^*_{ij} = \int_{\text{ray } i} \frac{\mathrm{d}\ell}{VQ} - \int_{\text{ray } j} \frac{\mathrm{d}\ell}{VQ}, \tag{2}$$

so that Δt^*_{ij} equals minus the fitted slope divided by π (Anderson and Hough, 1984; Quan and Harris, 1997). Spectra are computed over a signal window from 0.2 to 4.2 s after the P onset, truncated 0.5 s before the estimated S arrival and requiring at least 2 s of usable signal. The equally processed noise window spans -10 to -1 s relative to the P onset; the fit is restricted to frequencies at which both records exceed a signal-to-noise threshold. The candidate band is 1–12 Hz, restricted below the sampling-dependent Nyquist limit; at least 10 bins must meet amplitude SNR ≥ 3 at both receivers. All distinct pairs of available spectra within each event are fitted by ordinary least squares. Non-finite estimates and $|\Delta t^*| > 0.75$ s are rejected; there is no additional R^2 cutoff or uncertainty weighting. The retained signed differences include 47.1 % negative values, which are permissible because station ordering, not path length, defines their sign. A low spectral R^2 remains a limitation: many pairs share a record and cannot be treated as independent evidence. The inversion solves jointly for cell perturbations and one term per station,

$$\Delta t^*_{ij} - \Delta t^*_{0,ij} = \sum_k G_{ij,k} \, \delta q_k + \tau_i - \tau_j, \qquad G_{ij,k} = \frac{\ell^i_k - \ell^j_k}{V_k}, \tag{3}$$

where δq_k is the perturbation in Q_k^{-1}, V_k is the fixed recovered velocity, and $\Delta t^*_{0,ij}$ is the starting-model prediction including paths outside the active mask. The station term τ absorbs receiver, coupling, site and residual-instrument contributions (Eberhart-Phillips and Chadwick, 2002). Since τ_i and τ_j enter with opposite sign, only their differences are determined.

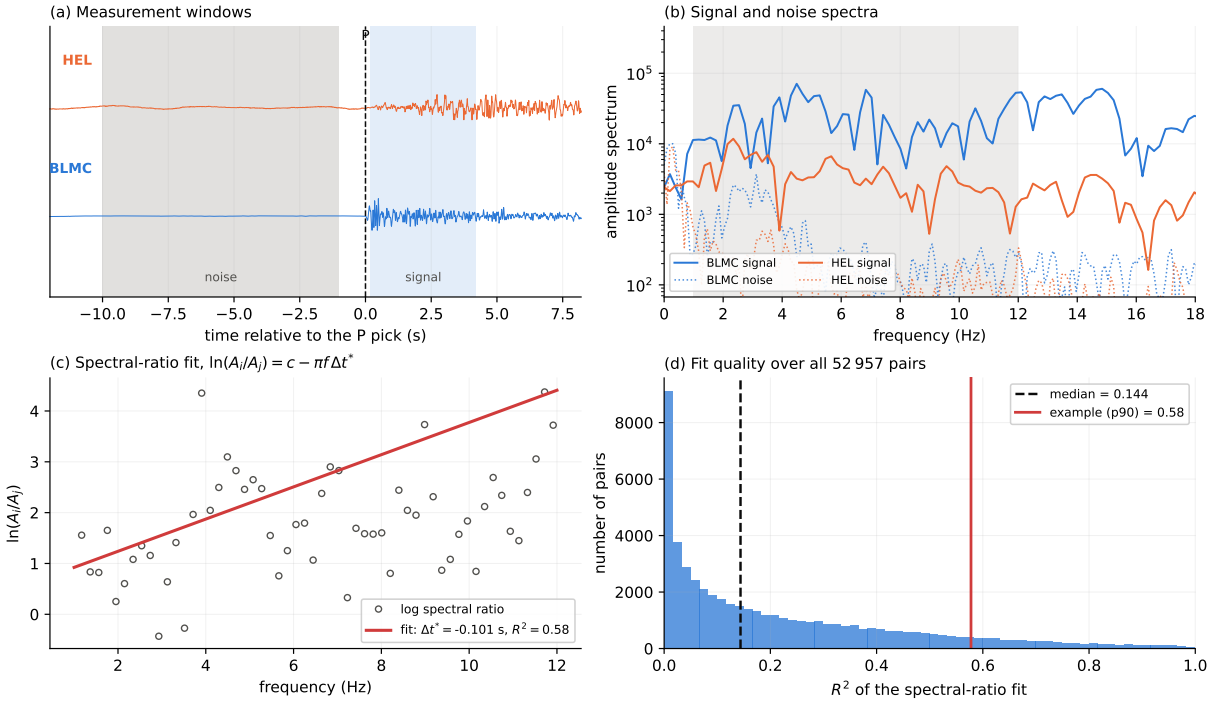


Figure 2: Measurement of a differential t^*. (a) Vertical-component records at two stations of one event, with the noise and signal windows used for the spectra; traces are normalised and offset. (b) Amplitude spectra of signal (solid) and noise (dotted) for both stations; the shaded band is the frequency range over which the ratio is fitted. (c) Logarithmic spectral ratio and its linear fit, whose slope gives Δt^* through Equation 2. (d) Distribution of the fit R^2 over all pairs, with the median and the example of panels (a)–(c) marked. The example was selected by declared percentile of R^2, not by inspection; the median of 0.144 indicates that most individual measurements are weak, which limits individual estimates. The number of pairs does not remove their shared-record dependence or establish unbiased attenuation.

3.4. Decomposition diagnostic

The archived statistic is normalised RMS reduction relative to a zero prediction, not normalised RMS reduction:

$$E_{\mathrm{RMS}} = 100\left(1 - \frac{\sqrt{N^{-1}\sum_n (d_n - \hat{d}_n)^2}}{\sqrt{N^{-1}\sum_n d_n^2}}\right). \tag{4}$$

It differs from R^2 and from the prediction-to-data variance ratio. Normalising by the data avoids dependence on a poor starting model, but does not measure how uniquely its components are determined. We partition the complete unconstrained linear prediction, including the starting contribution, as

$$\hat{d}_n = \underbrace{d_{0,n} + \sum_k G_{n,k}\delta q_k}_{c_n} + \underbrace{\tau_i - \tau_j}_{u_n}. \tag{5}$$

All component standard deviations and covariances use the same retained observations. They obey the exact centred identity

$$\mathrm{Var}(\hat{d}) = \mathrm{Var}(c) + \mathrm{Var}(u) + 2\,\mathrm{Cov}(c,u). \tag{6}$$

The two standard-deviation ratios indicate component inflation; values near unity alone do not establish good conditioning or independence. We therefore also report their correlation. Strong anticorrelation with

large component amplitudes is evidence of cancellation, consistent with a station–structure trade-off (Menke, 2018). It is a diagnostic, not a direct estimate of the operator's condition number. These diagnostics describe the unconstrained least-squares prediction, not forward predictions from the subsequently clipped model. The fit statistic and all moments in Table 1 use exactly the same retained observation vector for each campaign. Since the solution minimises an augmented, regularised objective, the data residual is not necessarily orthogonal to the prediction. The centred variance identity therefore does not imply an ordinary-least-squares relation between E_{RMS} and prediction variance. We also report $\mathrm{Corr}(\hat{d}, d)$.

To estimate sampling variability we draw 10000 bootstrap samples of events with replacement, retaining all pairs from each sampled event together (seed 20261005). We report the 2.5th and 97.5th percentiles of E_{RMS}. Expanded-domain comparisons share exactly the same pairs, so their contrasts use the same event draw in both configurations. The reference population is resampled separately. These intervals condition on the archived predictions: the models are not refitted. They account for within-event dependence, but not shared-station dependence across events, parameter uncertainty or systematic errors, and are not confidence intervals for individual tomographic values.

3.5. Ray-consistent parameterisation

Illumination is measured by tracing the rays of 8 000 accepted spectral-ratio pairs through the grid and counting cell hits. The pairs are selected by random sampling without replacement with seed 20260924. Rays are straight, and their geometric paths do not bend in either the initial or final velocity model; the latter enters only as a travel-time/attenuation weight. The inverted volume is then defined to contain the illumination rather than inherited from the nominal study area. The mask is changed in the original campaign comparison, and the sequential velocity recomputation also changes attenuation weights. The illumination audit counts nodes with at least one hit among the sampled pairs. This is the rounded, sample-scaled threshold $\max[1, \mathrm{round}(5 \times 8\,000/52\,914)] = 1$, distinct from the five-hit activation criterion applied to the complete inversion dataset. A sampled hit is evidence of geometric coverage, not sufficient evidence of resolution.

We additionally perform a controlled two-by-two attenuation experiment: each inherited or ray-domain active mask is combined with each saved inherited or updated velocity field. The grid, retained pairs, starting Q, quadrature and original regularisation, including station damping 0.15, remain fixed. Only the selected mask and velocity weights change. These are new unconstrained linear fits, not bounded inversions; their two diagonal cases reproduce archived residual RMS to within 10^{-7} s (maximum pointwise prediction difference below 10^{-4} s). Small floating-point summation differences change the LSMR stopping iteration without materially changing the fit.

3.6. Regularisation

Station damping λ_τ is swept over more than two orders of magnitude. The adopted value balances reduced component inflation against loss of normalised RMS reduction (Hansen, 1992; Menke, 2018). This is an empirical choice within the tested sweep, not proof of an optimum or complete elimination of non-uniqueness. Structure-only synthetic tests provide a separate check on leakage into station terms.

3.7. Resolution tests

Checkerboard patterns are imposed with sign $(-1)^{\lfloor i/n \rfloor + \lfloor j/n \rfloor + \lfloor k/n \rfloor}$ on grid indices, giving a declared wavelength of $2n$ grid spacings, and are repeated at 40, 80, 120, 160 km because a single scale cannot distinguish a resolved model from a smoothed one (Lévêque et al., 1993; Rawlinson and Spakman, 2016). Synthetic data are generated on the final geometry and inverted under the regularisation of the reported model. Two noise levels are used: a low level and a level equal to the adopted campaign's archived residual RMS before clipping (attenuation 0.0590 s; velocity 0.635 s). These are declared noise scenarios, not rigorous bounds on true resolution, because the residual contains unmodelled signal and the simulated noise does not represent shared-record errors. For the attenuation branch the synthetic data contain structure only, with $\tau \equiv 0$, so that the test additionally measures how much structure the free station terms absorb.

3.8. Model clipping and ceiling sensitivity

The archived implementation solves an unconstrained linear least-squares system and then clips Q^{-1} to the interval corresponding to $20 \leq Q \leq 300$. It does not solve a bound-constrained objective. Its stored prediction is evaluated from the unconstrained perturbation, before clipping. Changing the ceiling to 1500 therefore cannot alter that stored RMS when the linear system is unchanged. We evaluate both saved clipped models with the same straight-ray operator and station terms, and report this forward RMS separately. Ceiling saturation is retained as a flag for sensitivity to the numerical bound, but equal archived RMS values cannot demonstrate that the data fail to constrain the upper range of Q.

3.9. Benioff-surface reconstruction

Intermediate-depth hypocentres lying within active velocity cells serve as controls for a thin-plate-spline surface with smoothing parameter 25 in the kilometre coordinate system, evaluated inside the convex hull of the controls. Geometry is defined by seismicity alone; velocity and attenuation are sampled on the resulting surface by trilinear interpolation; active-mask membership uses the nearest grid node. The fine surface sampling describes hypocentral geometry, not finer tomographic resolution. Interpolation across a 20 km grid can mix active and inactive or clipped nodes, so attribute maps are descriptive rather than independent measurements at each surface node. Using Q to constrain the surface on which Q is subsequently read would be circular and is avoided. Local dip is obtained from the surface gradient; a single plane fitted to the whole surface averages steep and gentle segments and is reported only as an overall trend. The surface is compared with Slab2 (Hayes, 2018), as an external benchmark. Overlap between source catalogues has not been excluded, so statistical independence of the two surfaces is not assumed.

4. Results

4.1. Reference and expanded campaign comparison

Expanding the network increased the spectral-ratio dataset by more than an order of magnitude and the number of active attenuation cells from 442 to 777, yet normalised RMS reduction moved only

Table 1: Attenuation diagnostics on the same retained observations within each row. Panel (a) reports normalised RMS reduction and conditional event-bootstrap 95 % intervals; panel (b) gives component moments and correlations for the complete unconstrained prediction. Expanded rows share the same observation vector; the reference uses a different one. Cells denote active attenuation nodes.

(a) Configuration	Stations	Obs.	Cells	E_{RMS} (%)	95 % interval
Reference	14	4 216	442	12.1128	10.22–14.15
Expanded, inherited domain	56	52 914	777	12.5855	10.90–14.23
Ray-consistent domain	56	52 914	1 169	15.5116	14.10–16.99
Adopted ($\lambda_\tau = 4$)	56	52 914	1 169	15.0472	13.65–16.51

(b) Configuration	σ_c/σ_d	σ_u/σ_d	$\mathrm{Corr}(c, u)$	$\mathrm{Corr}(\hat{d}, d)$
Reference	1.1093	1.0788	-0.9188	0.4746
Expanded, inherited domain	3.8338	3.7366	-0.9895	0.4875
Ray-consistent domain	2.3972	2.3819	-0.9763	0.5323
Adopted ($\lambda_\tau = 4$)	0.9999	1.0346	-0.8680	0.5247

from 12.1 % to 12.6 % (Table 1). Conventional RMS reduction is 49.8 % relative to a starting residual RMS of 0.12281 s in the reference campaign, and 70.6 % relative to 0.20672 s in the expanded campaign. Their respective data RMS values are 0.07018 and 0.06941 s. These reductions use different starting predictions, generated by the respective three-dimensional and one-dimensional starting velocity fields. They measure improvement over different baselines and cannot by themselves compare goodness of fit across the campaigns.

The decomposition of Equation 5 resolves the discrepancy. In the reference configuration the structural and station contributions had ratios 1.11 and 1.08 relative to the data. After expansion these rose to 3.72 and 3.74. Their correlation changes from -0.919 in the reference campaign to -0.989 after expansion, confirming increased cancellation in the complete linear prediction.

4.2. Illumination and parameterisation

Tracing 8 000 pairs shows that 1 410 cells receive rays, of which 643 (46 %) lie outside the inherited domain and carry 29 % of all hits. Frozen at the one-dimensional starting model, their misfit can be accommodated only by the station terms, which is the behaviour the decomposition detects.

Re-parameterising over the illuminated volume (longitude $-75.10°$ to $-72.35°$, latitude 5.30° to 8.05°, depth to 140 km; 2 048 cells), retaining the spectral pairs and regularisation but recomputing velocity before attenuation, raised normalised RMS reduction to 15.5 %, reduced the trade-off ratios to 2.40 and 2.38, and increased active attenuation nodes to 1 169. The velocity branch has 907 active nodes and archived linear RMS 0.635 s. The reference has 365 active nodes, but its different data and starting field preclude interpreting the count increase alone as improved velocity accuracy.

The paired event-bootstrap increase from the inherited to the ray-domain campaign is 2.93 percentage points, with conditional 95 % interval 2.06 to 4.14. The interval excludes zero under this resampling

Table 2: Controlled attenuation fits on the same retained pairs, with original station damping. Only the active mask and saved velocity weights vary; fit statistics refer to unconstrained linear predictions.

Velocity weights	Active mask	E_{RMS} (%)	Residual RMS (s)
Inherited	Inherited	12.5855	0.06068
Inherited	Ray domain	15.5060	0.05865
Updated	Inherited	12.9083	0.06045
Updated	Ray domain	15.5116	0.05865

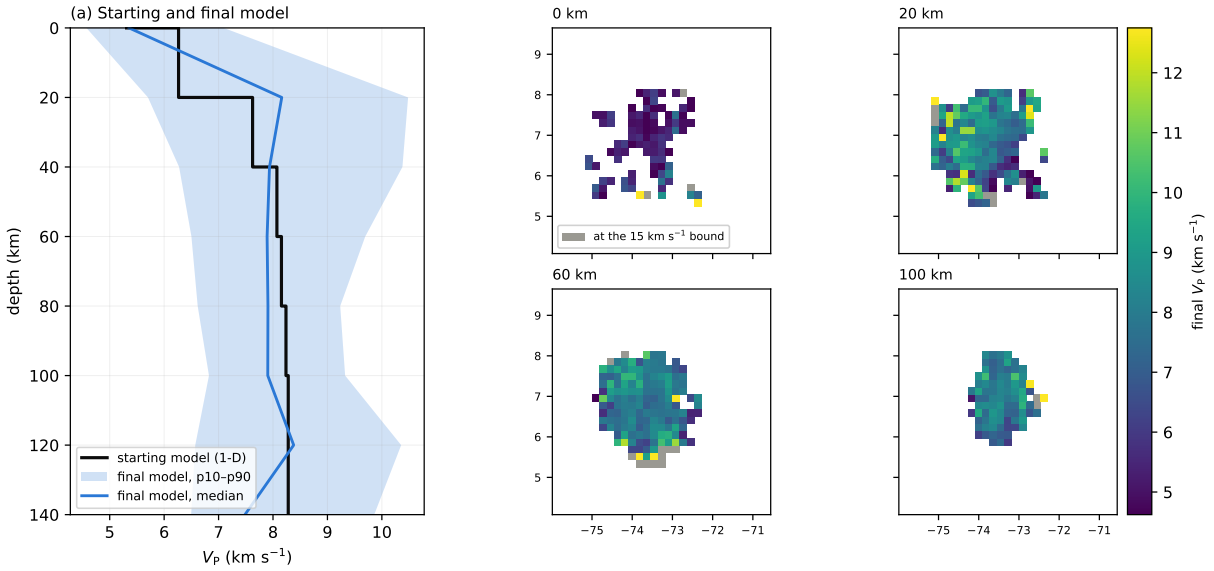


Figure 3: Starting and final P-wave velocity models. (a) The one-dimensional starting profile, with the median and the 10th–90th percentile range of the final model in each layer; the starting model is shown as a profile rather than as depth slices because it has no lateral structure. (b) Final model at four depths, restricted to active cells. Cells resting against the imposed upper bound on velocity (15 km s^{-1}) are shown in grey and excluded from the colour scale; they are a limit of the algorithm rather than a measurement, and are the velocity counterpart of the Q ceiling of Section 4.5.

scheme, while its scope is limited to the fixed-prediction comparison described above. The controlled experiment (Table 2) separates the mask and sequential velocity effects. At fixed inherited velocity, switching to the ray mask increases E_{RMS} by 2.9205 percentage points. Updating the velocity within that mask adds only 0.0056 points. The masking effect dominates this comparison, although the two factors interact and this result does not validate a bounded attenuation map.

4.3. Regularisation and the adopted model

The sweep reduces component inflation at limited cost in the linear fit: λ_τ can be raised from 0.15 to 4 with normalised RMS reduction falling only from 15.5 % to 15.0 %, while the trade-off ratios fall from 2.40 to 1.00 and from 2.38 to 1.03. Further damping reduces the normalised RMS reduction as well as component amplitudes; we retain $\lambda_\tau = 4$ as the adopted compromise (Fig. 4). The component correlation remains -0.868, compared with -0.976 at the original damping. The paired conditional bootstrap change in E_{RMS} is -0.46 percentage points (95 % interval -0.61 to -0.33); the reduction is small but consistent

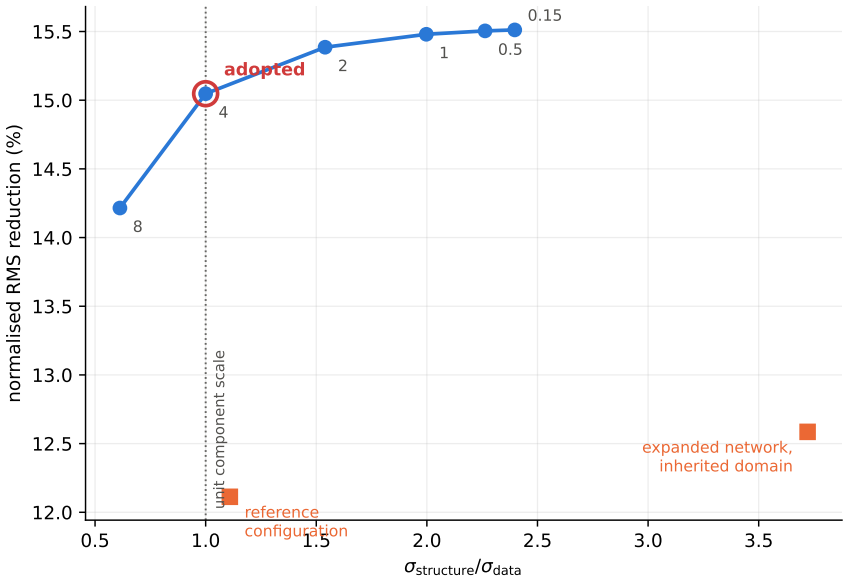


Figure 4: Trade-off between component inflation and linear fit. The abscissa is the standard deviation of the structural component of Equation 5 normalised by that of the data; the ordinate is normalised RMS reduction relative to the data. Points are labelled by λ_τ. The expanded network on the inherited domain lies far to the right, with components several times larger than the data and little normalised RMS reduction. The sweep traverses the curve leftwards; the compromise at $\lambda_\tau = 4$ is adopted. Ratios near unity do not establish independence of the components.

across event resamples of these fixed predictions. Anticorrelation is reduced, but the partition is still strongly coupled. Unity-scale ratios therefore do not justify calling the result well conditioned or uniquely resolved. Figure 5 maps the actual station terms. Damping can suppress genuine receiver response as well as compensating terms; these data provide no independent calibration that separates the two. The synthetic tests establish reduced spurious leakage only for their declared structure-only inputs.

4.4. Resolution

Checkerboard recovery is summarised in Table 3. The velocity branch recovers the imposed pattern almost completely at low noise ($r = 0.990$ at 160 km) but degrades sharply at the observed residual level ($r = 0.424$), with amplitude ratios exceeding unity, the expected behaviour when noise dominates. The attenuation branch is markedly less noise-sensitive, falling only from $r = 0.905$ to $r = 0.869$ at 160 km, and recovers the tested patterns at wavelengths of 80 km and longer on this fixed geometry. The 80 km pattern has four grid spacings per wavelength. Its recovery is a scale-dependent synthetic result, not a claim that arbitrary structures at that scale are identifiable or that clipped-model amplitudes are validated.

Because the synthetic data contain no station terms, any recovered τ is spurious. Under the adopted regularisation these have standard deviations of 0.0058–0.0071 s, one order of magnitude below the data. Repeating the test with the original $\lambda_\tau = 0.15$ raises them to 0.0163–0.0346 s while simultaneously lowering structural recovery at every wavelength (for example $r = 0.844$ versus $r = 0.869$ at 160 km). The adopted regularisation therefore improves both quantities, independently of the criterion by which it was chosen.

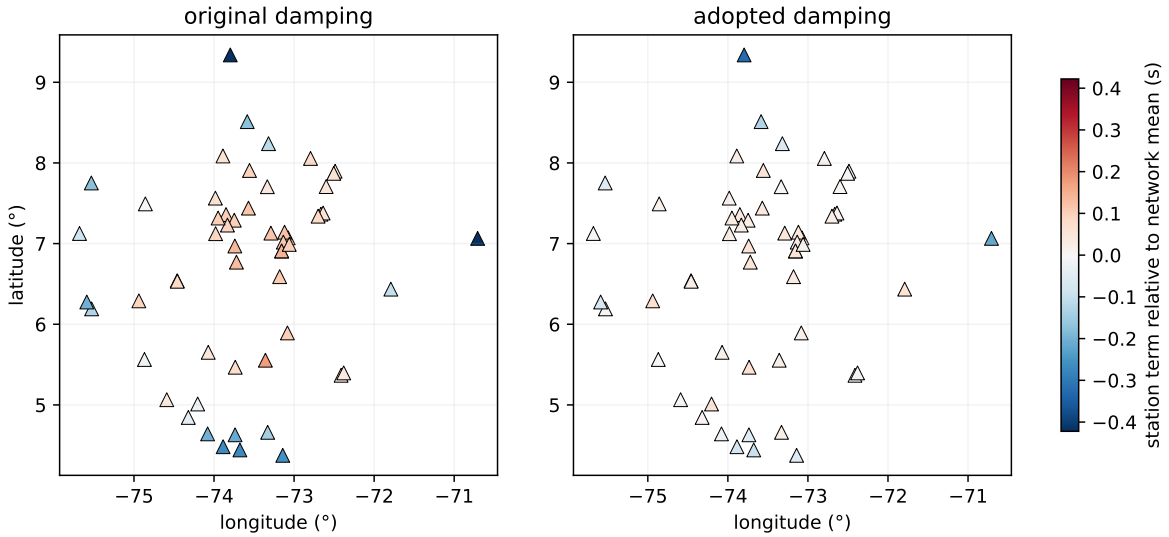


Figure 5: Station terms on the ray-consistent domain at original and adopted damping, expressed relative to the network mean on a common colour scale. Only station differences enter the observations. The reduced amplitudes may include suppression of genuine receiver response; no independent site calibration is available to separate that effect from compensation for path structure.

Table 3: Checkerboard recovery (correlation, and amplitude ratio in parentheses) for imposed perturbations of 5 % in slowness and 30 % in Q^{-1}. The low-noise and residual-noise columns describe two stated noise scenarios; they are not guaranteed upper and lower bounds on actual resolution.

	V_P		Q	
λ (km)	low noise	observed	low noise	observed
40	0.915 (0.86)	0.331 (1.10)	0.373 (0.42)	0.354 (0.45)
80	0.978 (0.96)	0.413 (1.29)	0.713 (0.65)	0.672 (0.65)
120	0.986 (0.96)	0.420 (1.22)	0.820 (0.78)	0.766 (0.76)
160	0.990 (0.97)	0.424 (1.12)	0.905 (0.86)	0.869 (0.84)

4.5. What the data determine about Q

The archived RMS difference 0.00000 s is expected from the implementation and is not an independent ceiling test. Forward evaluation of the saved clipped models instead yields 0.11641 s at $Q_{\max} = 300$ and 0.10882 s at $Q_{\max} = 1500$, compared with 0.0590 s for the unconstrained linear prediction. The reported linear fit therefore cannot be assigned to either bounded map. Both clipped-model residual RMS values exceed the data RMS 0.06941 s: their E_{RMS} values are -67.71 % and -56.77 %, respectively. Thus the displayed maps, combined with the archived station terms, fit worse than predicting zero differential attenuation. Velocity and station terms are held fixed in both forward evaluations; re-estimating the station terms after clipping could reduce these residuals. The two RMS values compare different clips of the same unconstrained solution and are not evidence favouring either ceiling.

At the original ceiling, 81 % of active nodes are clipped; 79 % of those nodes reach the higher ceiling when it is changed. The remaining 223 nodes (19 %) have median $Q = 158$ and 10th–90th percentiles

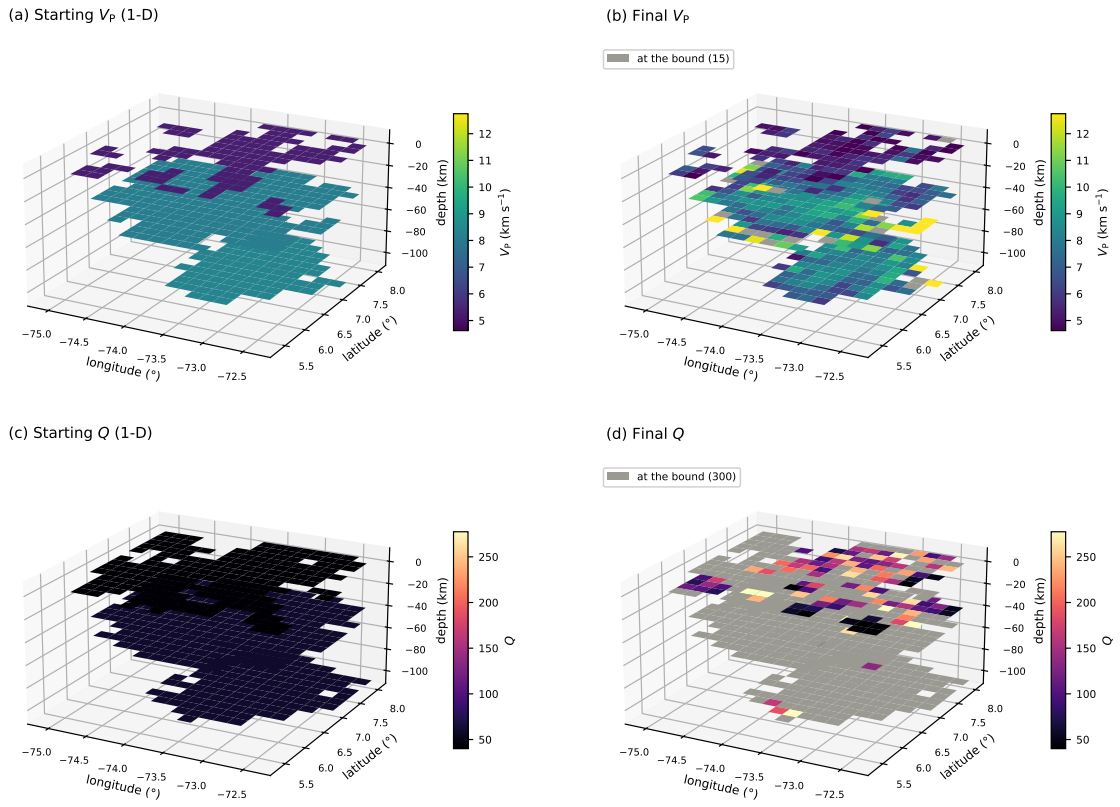


Figure 6: Starting and final models in isometric view, shown as horizontal planes at 0, 40 and 100 km and restricted to active cells. (a) and (c) are the one-dimensional starting models: lateral variation within each layer is zero to numerical precision for V_P, and Q takes only two values. Lateral departures from the starting models in (b) and (d) arise from the inversion, regularisation and subsequent clipping; they cannot all be attributed to data alone. In (b) and (d), cells resting against the imposed bounds on velocity and on Q are shown in grey and excluded from the colour scale; their extent in (d) is the visual counterpart of the ceiling test of Section 4.5.

64–260. Their median changes by 0.0 % because the same unconstrained solution is clipped differently, not because a constrained inversion independently recovered them. Being below the ceiling is necessary for considering a value, but does not establish that it is constrained by the data. The maps are therefore provisional descriptions of the clipped output. At the other bound, 0 active nodes (0.0 %) reach $Q = 20$, both with the original and the higher ceiling.

Below-ceiling nodes are not uniformly distributed (Fig. 7). The proportion at the ceiling is 55 % at the surface and 97 % at 100 km, so below-ceiling values are concentrated in the upper crust. The medians over below-ceiling nodes within each depth layer are 141 at the surface, 123 at 20 km and 157 at 40 km.

4.6. Benioff surface

The surface is constrained by 879 hypocentres and spans 59–150 km depth over 6 457 nodes, with a median distance of 5.6 km from any node to its nearest control. Median local dip is 43° (90th percentile 63°); the single-plane trend of 19° is reported only for completeness, as it averages across a curved surface. Compared with Slab2 over 6 457 nodes, the median signed difference is -2.2 km and the median absolute difference 8.2 km, with a 90th percentile of 28 km.

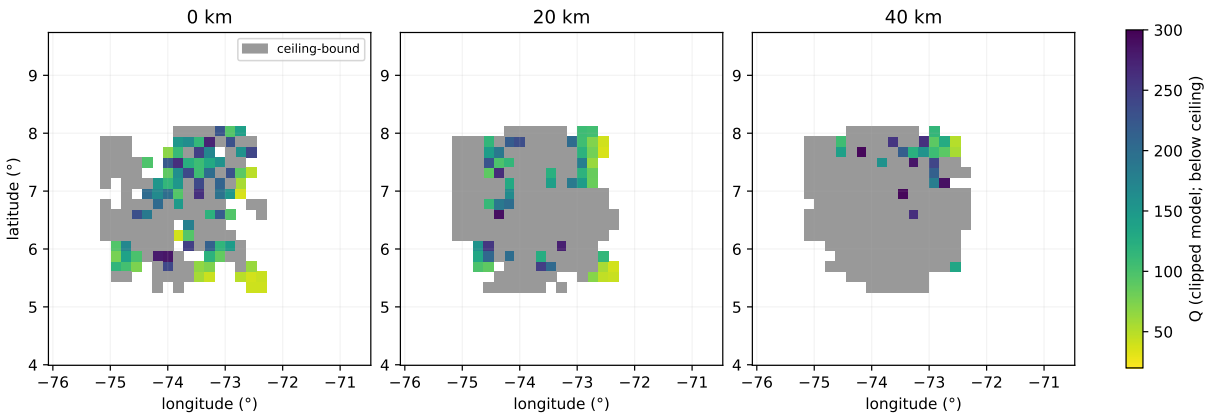


Figure 7: Horizontal slices of the saved clipped attenuation model. Ceiling-bound active nodes are grey and inactive nodes are blank. The colour scale includes only below-ceiling values. These are provisional maps: their forward RMS differs from the archived unconstrained fit, and absence of clipping alone does not establish data support.

5. Discussion

5.1. Illumination and parameterisation are distinct constraints

In this dataset, the expanded campaign on an inherited domain increases component inflation while producing little improvement in normalised linear RMS reduction. Domain expansion and stronger station damping reduce that inflation and improve structure-only synthetic recovery. The remaining anticorrelation shows that the two contributions are still coupled. Reporting their normalised amplitudes and covariance together is more informative than either the initial-model RMS reduction or a unity-scale amplitude criterion alone.

The result is a case study rather than a general argument against adding receivers. Additional stations may improve identifiability when they add independent path directions, when the domain already spans the illuminated volume, or when site response is independently constrained. Dense networks and different source distributions need not exhibit the same trade-off. Here the spectral pairs share records, and the straight-ray approximation and fixed hypocentres further restrict what can be inferred.

5.2. Clipping must be distinguished from constrained inference

Many nodes in the archived maps reach their numerical bounds, but the recorded predictions belong to an unconstrained linear solution. The ceiling comparison exposes a mismatch between the model displayed and the model whose residual is reported. A fraction of ceiling-bound nodes is still a useful diagnostic; however, unchanged pre-clipping RMS is not evidence that the data permit arbitrarily high Q. Neither the ceiling-bound nor the below-ceiling values can be labelled measurements on the basis of this test. A bound-constrained inversion, with predictions evaluated from its returned model and uncertainty or sensitivity tests for the bounds, is required for that interpretation.

5.3. Provisional attenuation structure of the basin

Figure 7 displays the clipped attenuation output with ceiling-bound nodes excluded from the colour scale. The medians over below-ceiling nodes are 141 at the surface, 123 at 20 km and 157 at 40 km. Thus

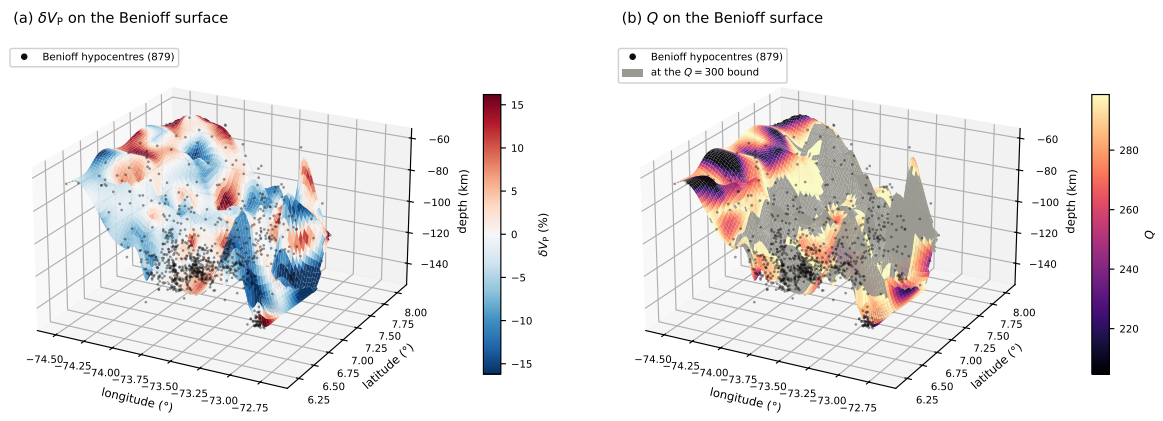


Figure 8: Benioff surface in isometric view, reconstructed from 879 intermediate-depth hypocentres (black dots), with (a) the velocity anomaly and (b) Q sampled on it as attributes. The geometry is defined by seismicity alone; neither field enters the surface fit. In (b) the nodes resting against the imposed Q ceiling are shown in grey and excluded from the colour scale, following Section 4.5; they cover a large part of the surface, so no quantitative attenuation interpretation is offered at intermediate depth. Trilinear attribute interpolation does not increase tomographic resolution.

the quoted range 123–157 describes two layer medians, not the range of individual values or the global median. The global below-ceiling median is 158, with a much broader 10th–90th percentile interval 64–260.

Sedimentary basin fill and fluid-sensitive attenuation provide geological hypotheses for these patterns (Mavko et al., 2009; Biot, 1956; Quan and Harris, 1997; Mojica and Franco, 1990; Alcárcel and Gómez, 2019; Londoño et al., 2022; Castelblanco et al., 2025). The present clipping audit, low spectral-fit quality and residual station–structure coupling prevent validation of that interpretation. We do not infer a fluid distribution or quantitative slab attenuation from these maps (Eberhart-Phillips and Chadwick, 2002; Jin et al., 2021).

5.4. Benioff geometry and velocity structure

The reconstructed surface agrees with Slab2 to 8.2 km in median absolute depth, providing a descriptive comparison with the external model (Hayes, 2018). We have not evaluated a regional Slab2 uncertainty field or propagated hypocentral errors, so this difference is not claimed to lie within a quantified uncertainty interval. Median local dip of 43° is consistent with the intermediate-depth segment beneath northeastern Colombia inferred from regional seismicity (Zarifi et al., 2007; Sepúlveda-Jaimes and Cabrera-Zambrano, 2018; Syracuse et al., 2016). The velocity field sampled on the surface shows a localised low-velocity anomaly within the Bucaramanga nest volume relative to its surroundings. Interpretation at this scale is restricted by the velocity resolution of Table 3: at the observed noise level the velocity branch recovers only $r \approx 0.424$ even at 160 km, so we regard the displayed anomaly as tentative and do not interpret its amplitude or sign as independently validated.

5.5. Limitations

The straight-ray operator and fixed hypocentres make this a linearised audit rather than a fully non-linear tomography solution; finite-frequency or eikonal modelling and relocation would change the kernels (Sethian, 1996; Zhao, 2005; Vavryčuk, 2012; Thurber, 1992). Velocity and attenuation maps are clipped after unconstrained optimisation, and the fit reported for that optimisation is not their forward fit. Checkerboards use the same operator as the inversion and therefore test recovery within that discretisation, not robustness to ray bending, model error or clipping. Independent Gaussian noise at the residual RMS does not reproduce record sharing or correlated spectral errors, and the two noise scenarios are not rigorous bounds on true resolution. No uncertainty weighting, held-out prediction or independent station-response calibration is available here. The fine Benioff surface grid likewise does not justify fine-scale interpretation of interpolated tomographic fields. These limitations prevent treating the clipped Q values as quantitative measurements, while leaving the archived linear trade-off comparison and its synthetic leakage diagnostics informative.

6. Conclusions

1. Increasing the receiver set from 14 to 56 stations and accepted spectral pairs from 4 216 to 52 914 accompanies a change in normalised RMS reduction only from 12.1 % to 12.6 % while increasing cancellation between the structural and station components. Starting velocity structure and computational extent also differ, so this comparison cannot isolate the effect of receiver expansion.
2. The inherited domain omits 46 % of nodes hit in the sampled pairs, carrying 29 % of ray hits. The paired event-bootstrap gain is 2.93 percentage points (95 % interval 2.06 to 4.14); at fixed inherited velocity the mask alone adds 2.9205 points to the linear fit. Re-parameterisation and stronger station damping reduce component inflation and improve structure-only checkerboard recovery, but the adopted component correlation of -0.868 shows that substantial coupling remains.
3. The archived RMS is evaluated before clipping. Forward RMS of the clipped attenuation model is 0.11641 s, changing to 0.10882 s at the higher ceiling. The earlier equality of archived RMS therefore cannot demonstrate upper-bound non-identifiability. Below-ceiling values are provisional output, not validated measurements.
4. A Benioff surface from 879 hypocentres differs from Slab2 by a median absolute 8.2 km. Its fine surface sampling does not improve the resolution of the interpolated tomographic attributes.

The operational recommendations are to report the covariance of structural and station contributions on the retained observations, to avoid using archived RMS invariance as evidence of bound non-identifiability, and to recompute forward predictions from the bounded model actually displayed. These checks precede quantitative interpretation of attenuation values.

CRediT authorship contribution statement

Rómulo Sandoval Flórez: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Francisco H. Cabrera Zambrano: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review and editing, Visualization. **Claudia Patricia Parra Medina:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review and editing, Visualization.

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Declaration of competing interests

The authors declare no competing interests.

Data Availability

Earthquake catalogues, phase picks and waveforms are distributed by the Servicio Geológico Colombiano. Slab2 is distributed by the U.S. Geological Survey (Hayes, 2018). The processed campaign tables, saved models, numerical configuration, covariance diagnostics, conditional event-bootstrap outputs, and clipped-model forward-prediction audit are provided in Supplementary Material S1. The code and processed research data are archived as version 1.0.2 at `https://github.com/romusan/vp-q-tomography-audit/tree/v1.0.2` (Sandoval Flórez et al., 2026). The archive supports the audit from saved processed inputs; it does not include raw waveforms or constitute complete reproduction from waveform acquisition through spectral preprocessing. Regeneration of the separate checkerboard and Benioff experiments is outside the verified scope; their archived summaries are included.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Codex to assist with language revision. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

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